

Answer Key — Biology Grade 11

Sub-Strand 1.3: Taxonomy II • Lesson 7: Dichotomous keys for animals and viruses

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
C	B	D	C	C	D	D	A

USE AFTER: PREDICT PHASE — Predict: message to a visitor from Kisumu (what kind of statements would work best)

Q1. A visitor from Kisumu has never seen the animals at your school. You can only send a written message so that the visitor names the same animal as you. Which statement would work best in the message?

- A. It is a beautiful animal.
- B. It is a big animal.
- C. It has six jointed legs.**
- D. It is a useful animal for food.

Correct answer: C. Six jointed legs is a feature anyone can see and count, so everyone reads it the same way. Big, beautiful and useful are opinions that differ between people.

USE AFTER: OBSERVE PHASE — Part A: swapping keys and identifying mystery animals

Q2. A group builds this part of a dichotomous key. 1a. Animal has a backbone: go to step 2. 1b. Animal has no backbone: go to step 3. 2a. Feathers present: chicken. 2b. Feathers absent, hair present: goat. 3a. Six jointed legs present: grasshopper. 3b. No legs, body covered by a hard shell: snail. A tester has an animal with no backbone and six jointed legs. Which animal does the key lead to?

- A. Goat
- B. Grasshopper**
- C. Chicken
- D. Snail

Correct answer: B. No backbone leads to step 3, and six jointed legs leads to grasshopper. Following one choice at each step gives one answer.

USE AFTER: OBSERVE PHASE — Part A: group records where the key failed or was ambiguous and fixes it

Q3. A group wrote this couplet: 'Animal is small' / 'Animal is big'. After swapping keys, the testing group got different answers from the authors. What is the main problem with this couplet?

- A. Animals never differ in size, so the feature is useless.
- B. The couplet should name the animal in the first statement.
- C. A couplet must have three choices, not two.
- D. Two people can disagree about what small and big mean, because the statements are not measurable or countable.**

Correct answer: D. Small and big depend on what each person compares with. A good couplet uses observable, countable features, such as a backbone or the number of legs, so every user reaches the same answer.

USE AFTER: OBSERVE PHASE — Part A: compare with the prepared key and sort animals into Chordata or Arthropoda

Q4. Learners sort animals into Phylum Chordata or Phylum Arthropoda. Which list contains only Arthropoda?

- A. Chicken, lizard, grasshopper
- B. Butterfly, earthworm, grasshopper
- C. Mosquito, butterfly, grasshopper**
- D. Mosquito, butterfly, snail

Correct answer: C. Mosquito, butterfly and grasshopper are insects with jointed legs, so they are Arthropoda. The snail and earthworm are neither Arthropoda nor Chordata. Chicken and lizard are Chordata.

USE AFTER: OBSERVE PHASE — Part B: virus diagrams and reading, table of virus characteristics

Q5. Which statement correctly describes a virus?

- A. It has a nucleus and mitochondria, like the cells of a goat.
- B. It is a very small bacterium that divides on its own in soil.
- C. It is a non-cellular particle with a protein coat around DNA or RNA, and it reproduces only inside a host cell.**
- D. It carries out respiration and growth by itself outside a host.

Correct answer: C. Viruses have no cell structure and no nucleus or mitochondria. They have genetic material (DNA or RNA) inside a protein coat and can reproduce only inside a host cell.

USE AFTER: EXPLAIN PHASE — CER: 'Is a virus a living organism?'

Q6. A learner claims that a virus is not a living organism. Which piece of evidence best supports this claim?

- A. A virus has genetic material, DNA or RNA.
- B. A virus can cause disease in maize and in people.
- C. A virus is very small.
- D. A virus has no cell structure and cannot reproduce or carry out metabolism outside a host cell.**

Correct answer: D. Living things show characteristics such as reproduction and metabolism on their own. A virus has no cells and depends on a host cell for these. Small size, causing disease and having genetic material do not show that something is non-living.

USE AFTER: EXPLAIN PHASE — Teacher probe: correct the misconceptions 'viruses are tiny bacteria' and 'antibiotics kill viruses'

Q7. A health worker says that antibiotics will not cure measles. What is the correct reason?

- A. Measles is caused by a very large bacterium that resists antibiotics.
- B. Antibiotics only work on diseases of plants.
- C. Measles is not caused by any organism or particle.
- D. Measles is caused by a virus, which is not a bacterium, and antibiotics work against bacteria.**

Correct answer: D. Viruses are non-cellular particles that live and reproduce inside host cells, and they are not bacteria. Antibiotics target bacteria, so vaccines are used to prevent virus diseases like measles.

USE AFTER: MODEL BUILDING PHASE — Revise the class model: add a box for viruses showing their importance in health, agriculture and biotechnology

Q8. Learners add a box for viruses to the class model. Which example shows a useful (beneficial) importance of viruses?

- A. Bacteriophages being used to attack harmful bacteria**
- B. Cassava mosaic disease reducing cassava yields on Kenyan farms
- C. Foot and mouth disease in cattle
- D. Newcastle disease killing chickens

Correct answer: A. Bacteriophages are viruses that infect and destroy bacteria, so they can be used against harmful bacteria. The other three examples are harmful viral diseases of crops and livestock.

